

Campbell Downs

West Lafayette, IN | Brentwood, TN

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Education

Purdue University

Expected May 2029

Bachelor of Science in Aerospace Engineering; Minor in Product Lifecycle Management

West Lafayette, IN

- **GPA: 3.31** | U.S. Citizen
- **Relevant Coursework:** Aerospace Design, Aeromechanics I, Thermodynamics I, Digital Manufacturing, Simulation & Visualization, Differential Equations

Technical Skills

CAD / PLM: SolidWorks, Siemens NX, Teamcenter (PDM), parametric modeling, assemblies, ECO workflows
Analysis: SolidWorks Simulation, linear static FEA, fatigue analysis, transient thermal analysis, design optimization
Programming / Data: MATLAB, Python, C, C#, JavaScript, HTML, Microsoft Excel
Manufacturing: 3D printing, additive manufacturing, introductory CNC machining, general shop tools

Certifications

Certified **SOLIDWORKS Associate (CSWA)** – Dassault Systèmes, September 2026

Experience & Engineering Involvement

Handshake AI

May 2026 – Present

AI Fellow (Independent Contractor)

Remote

- Completed **350+ contracted hours** evaluating AI outputs and annotating video data for multimodal action-recognition and tracking models.
- Averaged approximately **4/5** on quality assessments, advanced twice to higher work-access tiers, and performed peer quality-control reviews.

Purdue Orbital

September 2026 – Present

Launch Structures Team Member

West Lafayette, IN

- Supporting the **HAPSIS II mission** in developing lighter, more robust launch and landing structures to improve survivability and reduce helium lift costs.

Nashville Adventure Science Center

2025 – 2026

STEM Education Volunteer

Nashville, TN

- Contributed **40+ volunteer hours** supporting STEM education, science outreach, and visitor engagement.

Selected Engineering Projects | Portfolio

Project ECHO – Endurance-Optimized Multirotor | Python, MATLAB, CAD, Systems Optimization 2026 – Present

- Developing a lower-cost agricultural imaging drone optimized for **endurance, system cost, and usable image quality**.
- Evaluating motors, propellers, ESCs, batteries, and imaging hardware through component trade studies and system-level optimization.

Multi-Station Fitness Rack | SolidWorks, Structural FEA, Fatigue Analysis

2026

- Designed an A36 steel fitness rack for a **250 lb user load** and validated four structural load cases using static FEA and fatigue analysis.
- Redesigned critical members based on simulation results until the assembly satisfied defined structural and fatigue requirements.

Bicycle Horn Mount Optimization | SolidWorks, Parametric & Topology Optimization

2026

- Evaluated **270 parametric configurations** of a Nylon 6/10 mount across five variables for mass, stiffness, and factor of safety.
- Combined parametric and topology optimization to reduce mass by approximately **30%** while maintaining FOS above 100.

Interests

Golf, strength training, baseball, and hiking